

MyGarage RAG System

VECTOR KNOWLEDGE BASE DOCUMENTATION & ARCHITECTURE GUIDE

Project Overview

MyGarage is a Retrieval-Augmented Generation (RAG) system built for automotive diagnosis and maintenance. It transforms a structured car knowledge base into a searchable vector database, enabling an AI assistant to accurately answer questions about car symptoms, repairs, and maintenance schedules.

Key Features & Functionality

- **Knowledge Chunking:** Segments 169 detailed automotive knowledge entries from a structured `car_final.txt` source file.
- **Vector Embeddings:** Converts each text chunk into a dense 1024-dimensional vector using the `mxbai-embed-large` model.
- **Database Storage:** Stores raw text data along with high-dimensional vectors in a PostgreSQL 18 database utilizing the `pgvector` extension.
- **Efficient Indexing:** Builds a Hierarchical Navigable Small World (HNSW) index to provide rapid approximate nearest-neighbor (ANN) search execution.
- **Advanced Ingestion & Retrieval Pipeline:** Performs vector similarity searches at query time to retrieve the top 10 candidates based on L2 distance, followed by CrossEncoder reranking to pass the top 3 optimal context blocks to the LLM.

File Structure

File	Role	Key Functions / Notes
<code>chunking_functions.py</code>	Core chunking logic	Contains <code>embed()</code> , <code>create_chunk()</code> , <code>insert_chunk()</code> , and <code>create_hnsw_index()</code> .
<code>chunks_bd.py</code>	Database model & config	Handles SQLAlchemy ORM mapping, <code>pgvector</code> Column declaration, and the DB engine setup.
<code>search_functions.py</code>	Search & rerank pipeline	Manages <code>search_function()</code> , <code>rerank_results()</code> , and the main <code>retrieve_content()</code> logic.
<code>chunking.ipynb</code>	Execution notebook	Orchestrates the entire end-to-end data ingestion and vector indexing pipeline.
<code>car_final.txt</code>	Knowledge base source	Contains 169 car maintenance and diagnostic entries, separated explicitly by <code>[SEP]</code> tokens.

Database Schema (Table: chunk)

Column	Type	Description
id	INTEGER (PK)	Auto-incrementing primary key.
title	TEXT	The extracted first line of the knowledge entry (e.g., # Worn Brake Pads).
content	TEXT	The full narrative text content belonging to the diagnostic chunk.
embedding	VECTOR(1024)	Dense vector representation generated via the mxbai-embed-large model.

Full Pipeline Step-by-Step

Step	Function	Description	Technology
1	extract_content()	Reads raw structured text from the source knowledge base file.	Python I/O
2	split('[SEP]')	Splits text into separate semantic entries based on delimiters.	Standard String API
3	embed()	Generates a 1024-dimensional dense vector for each text chunk.	mxbai-embed-large
4	insert_chunk()	Stores the textual components alongside the vector embedding in the DB.	pgvector
5	create_hnsw_index()	Constructs a high-performance HNSW index for rapid ANN queries.	pgvector HNSW
6	search_function()	Executes L2 distance similarity searching to extract the top 10 matches.	SQLAlchemy
7	rerank_results()	Re-ranks the candidate list using a CrossEncoder to filter the top 3 entries.	BAAI/bge-reranker-base
8	retrieve_content()	Delivers the final structured, high-relevance context to the LLM agent.	RAG Pipeline Core

Technology Stack

Layer	Technology	Version / Implementation Notes
Database Engine	PostgreSQL	Version 18, acting as the primary relational data store.
Vector Extension	pgvector	Enables native VECTOR(1024) column storage and operations.
Vector Indexing	HNSW Index	Configured with L2 distance operations (vector_l2_ops).
Embedding Model	mxbai-embed-large	Local execution generating 1024 dimensions via Ollama.
Reranking Model	BAAI/bge-reranker-base	CrossEncoder implementation executing on a CUDA-accelerated device.
ORM / Driver	SQLAlchemy	Paired with the psycopg (v3) database driver.
Runtime Environment	Python 3.10	Orchestrated via Jupyter Notebooks and structured python modules.

Knowledge Base Format (car_final.txt)

Every database entry originates from the `car_final.txt` text file. Chunks are separated using the `[SEP]` token. The first line automatically serves as the segment title, while the subsequent text block forms the complete chunk content.

```
# Worn Brake Pads
Sound: Squealing, Screeching, Grinding
What Broke: Brake pads are too thin.
How to Recognize While Driving: Noise only when pressing brake pedal, Grinding means metal touching metal
Main Causes: Wear, Age
Is It Dangerous?: High Risk
Easy or Garage?: Garage recommended
Typical Fix: Replace brake pads (and rotors if scored)
```

Database Inspection & Verification (Live State Verification)

Below are active query inspection panel layouts verified directly from the PostgreSQL 18 client dashboard, confirming the 169 imported chunks and vector formatting.

pgAdmin IV - MyGarage/postgres@PostgreSQL 18

```
SELECT * FROM chunk;
```

Data Output | Messages | Notifications Showing rows: 1 to 169

id [PK]	title [text]	content [text]
1	# Worn Brake Pads	# Worn Brake Pads Sound: Squealing, Screeching, Grindin...
2	# Warped Brake Discs (Rotors)	# Warped Brake Discs (Rotors) Sound: Grinding, Pulsing ...
3	# Wheel Misalignment	# Wheel Misalignment Sound: Usually no loud sound, Tire...
4	# Tire Balancing Problem	# Tire Balancing Problem Sound: Humming, Vibration nois...
5	# Bad CV Joint (Cardan / Axle Joint)	# Bad CV Joint (Cardan / Axle Joint) Sound: Clicking, P...
6	# Bad Steering Rack	# Bad Steering Rack (Rack and Pinion) Sound: Clunking, ...
7	# Loose Tie Rod Ends	# Loose Tie Rod Ends Sound: Clunking, Knocking What Bro...

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```
SELECT "content", "embedding" FROM "chunk";
```

Data Output | Messages | Notifications Showing rows: 1 to 169

content [text]	embedding [vector]
# Worn Brake Pads Sound: Squealing, Screeching, Grin...	[-0.04682187, 0.03936265, 0.0124513, -0.01982...
# Warped Brake Discs (Rotors) Sound: Grinding, Pulsi...	[-0.03602896, 0.02629002, -0.0048123, 0.05129...
# Wheel Misalignment Sound: Usually no loud sound, T...	[-0.01236811, 0.04105354, 0.0891241, -0.03141...
# Tire Balancing Problem Sound: Humming, Vibration n...	[-0.04603844, 0.03114100, -0.0154812, 0.02294...
# Bad CV Joint (Cardan / Axle Joint) Sound: Clicking...	[-0.032168265, 0.0541450, 0.0014812, -0.06718...